OPEN ACCESS

Four distinct patterns of anterior cruciate ligament injury in women's professional football (soccer): a systematic video analysis of 37 match injuries

Leonard Achenbach ,[1] Hendrik Bloch,[2] Christian Klein ,[2] Theresa Damm,[3] Matthias Obinger,[4] Maximilian Rudert,[1] Werner Krutsch,[5,6] Dominik Szymski[5]

[1]Department of Orthopedics, Julius Maximilians University Würzburg, Wurzburg, Germany
[2]Department of Sports Injury Prevention, VBG, German Statutory Accident Insurance for the Administrative Sector, Bielefeld, Germany
[3]Faculty of Medicine, University Würzburg, Würzburg, Germany
[4]University of Cooperative Education for Sports and Health, Baunatal, Germany
[5]Department of Traumatology, University Clinic Regensburg, Regensburg, Germany
[6]SportDocsFranken, Nuremberg, Germany

Correspondence to
Dr Leonard Achenbach, Department of Orthopedics, Julius Maximilians University Würzburg, Würzburg, Germany; leonardachenbach@gmail.com

Accepted 17 April 2024
Published Online First 29 April 2024

ABSTRACT

Background To identify mechanisms and patterns of anterior cruciate ligament (ACL) injury in adult women's professional football by means of video match analysis.

Methods ACL match injuries sustained in Germany's first women's league during the 2016–2017 to 2022–2023 seasons were prospectively analysed by three expert raters using a standardised observation form. Epidemiological and injury data, as well as the medical history of ACL tears, were obtained from media reports and the statutory accident insurance for professional athletes.

Results Thirty-seven ACL injuries sustained in official football matches were included in the video analysis, of which 24 (65%) had associated knee injuries, mainly meniscus and collateral ligament injuries. According to the categorised contact mechanisms, 17 (46%) were non-contact injuries, 14 indirect contact injuries (38%) and six direct contact injuries (16%). Of the 17 non-contact injuries, seven (41%) occurred during the first 15 min of the match. Contact mechanisms did not differ between primary and secondary ACL injuries to the same or the contralateral side. Most injuries (80%) of field players occurred during horizontal movements such as sprinting (n=9, 26%), change-of-direction manoeuvres (n=7, 19%), stopping (n=5, 14%) and lunging (n=5, 14%). Four distinct repetitive patterns of ACL match injuries were identified: (1) non-contact 'pressing ACL injury' (n=9), (2) indirect contact 'parallel sprinting and tackling ACL injury' (n=7), (3) direct contact 'knee-to-knee ACL injury' (n=6) and (4) non-contact 'landing ACL injury' (n=4).

Conclusion Most of the identified patterns of ACL injuries in women's professional football have great potential for prevention.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ The risk of anterior cruciate ligament (ACL) injury in professional women's football is high.
⇒ Little is known about frequent injury patterns in this high-risk population.

WHAT THIS STUDY ADDS

⇒ The four typical injury patterns for ACL injury in women's professional football were (1) non-contact 'pressing ACL injury', (2) indirect contact 'parallel sprinting and tackling ACL injury', (3) direct contact 'knee-to-knee ACL injury' and (4) non-contact 'landing ACL injury'.
⇒ A high proportion (80%) of ACL injuries were sustained during horizontal (rather than vertical) movements, and 41% of non-contact injuries occurred in the first 15 min of play.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ The results provide insight into how ACL injuries occur in women's professional football and identify typical situational patterns that should be considered in risk-reduction strategies.
⇒ Timing appears to be a contributing factor in non-contact injuries, suggesting a high potential for prevention through adequate match preparation, neuromuscular activation and load management.
⇒ Preventive exercises should consider improving lateral and rotational trunk stability during one-legged stance during sprinting and upper-body contact with rotation. In addition, alternative tackling and landing techniques could be taught for these at-risk situations.

BACKGROUND

Professional female football players are at high risk of an anterior cruciate ligament (ACL) injury.[1] In the last two decades, the incidence of ACL injuries in girls has seen a major increase,[2–4] with incidences increasing from 0.06 per 1000 hours[5] to 0.1–0.2 per 1000 hours.[6–8] Female athletes participating in intermittent team sports (including football) are reported to have a 2–6 times higher risk of injury than male athletes.[9 10] The annual prevalence of ACL injuries in female players ranges from 0.5% to 6.0%. For both male and female players, studies have consistently shown a 7–65 times higher incidence of ACL injury during match play.[10] ACL reconstruction surgery is often required, followed by a long rehabilitation period. A 2016 systematic review showed a rate of 26% of a second ACL injury to either knee after return-to-play in athletes under the age of 25.[11] ACL injuries can not only lead to long-term physical impairment, reduced athletic performance and premature career ends but are also associated with an increased risk of osteoarthritis.[12 13]

Video analysis plays a vital role in sports injury prevention research by allowing the detailed examination of biomechanics, movement patterns and player behaviour.[14] This detailed analysis provides valuable insights for the development of targeted interventions and strategies to reduce the incidence of sports injuries.[15 16]

Check for updates

To cite: Achenbach L, Bloch H, Klein C, *et al. Br J Sports Med* 2024;**58**:709–716.

Previous video analysis studies investigating ACL injury mechanisms in football players primarily focused on men[17 18] or mixed cohorts. In the mixed cohort study of non-professional athletes, Brophy *et al* identified a defensive pressing pattern as a common cause of ACL injuries in women.[19] This finding was supported by Lucarno *et al*[20] in a female-only study that found the defensive pressing pattern to be the leading cause of ACL injuries among three main injury patterns.[20]

Despite these two studies, essential information is still lacking in women's football on ACL injury mechanisms. Indeed, a more detailed knowledge of injury situations in women's professional football may help establish more targeted injury prevention measures.[14] Thus, the purpose of this study was to identify mechanisms and patterns of ACL injuries in women's professional football in a first-league setting by using video match analysis, with particular emphasis on differences between first and secondary ACL injuries, differences in associated knee injuries and the role of foul play, timing of injury and substitution.

METHODS

This prospective study was conducted from the 2016–2017 season to the 2022–2023 season. The study included all ACL injuries sustained during official matches of a football team of the first women's national league in Germany (1. Frauen Bundesliga). Only female players who sustained an ACL injury during an official match in the first national league, German Cup or Champions League were included. Injuries sustained during matches with the national team were systematically excluded. Injuries and contact mechanisms were defined by video match analysis as previously described by our study group[21–23] and according to current guidelines.[24] Briefly, a contact injury was defined as any injury due to external forces directly at or adjacent to the injured body site, either by contact with another player or with an object such as the goal or the ball, immediately before or at the same time as the suspected injury. Indirect contact injuries were defined as any injury due to external forces that did not directly cause the injury but affected the natural movement process and indirectly led to the injury. Non-contact injuries were defined as any injury that did not involve contact with another player or object.

Inclusion and exclusion criteria, the procedure of video production, the provision of video footage, data collection and video analysis were identical to those described in previous literature reports.[21–23 25] In short, this prospective cohort study is based on the nationwide ACL registry in German football, to which all professional football clubs in the German first division were invited to report all new ACL injuries of their players.[26] In addition, national media reports on professional football players were systematically screened on a weekly basis in various German online and print media. After the registration of a new ACL injury, either through direct notification by the players and clubs or through media reports, a standardised questionnaire was sent to the injured player directly or via a team member. These medical data were entered into the database of the national insurance company, Verwaltungs-Berufsgenossenschaft (VBG). The VBG receives all time-loss injury reports from the team physicians. The data from the media reports were systematically compared with the physician's reports and, if necessary, completed. Associated knee injuries were systematically assessed from the team physician's reports.

Standardised video analysis

Three expert raters with experience in professional football (LA, TD and CK) reviewed and classified each injury-causing event according to the applied observation form, which had been used in previous similar studies.[21 27 28] A detailed description of the video analysis process is given elsewhere.[21–23] In short, we used video footage provided by DieLigen GmbH, publicly available video footage from the German broadcasters ARD and ZDF and video matches uploaded from the social media channels of the respective football clubs. The footage was captured by at least one camera in the stands of each football stadium. Information on the approximate time of the injury was taken from the website of the German football magazine 'kicker' (https://www. kicker.de) and, if necessary, from an additional Google search (name of player AND injur* AND name of opposing team). Each clearly identified injury situation was cut to a sequence of approximately 10–15 s before and approximately 5–10 s after the injury event to assess the specific match situation, the injury circumstances and the referee's decision. In addition, all available replays from the television broadcast (in slow motion and from different perspectives) were added to the sequence. In the next step, LA, TD and CK independently assessed the video footage using the described observation form. First, LA and TD analysed the videos and held an initial consensus meeting. A third rater (CK) was added subsequently. In a subsequent consensus meeting, the assessments were compared, and deviating assessments were discussed and harmonised by consensus. Frequent repetitive injury patterns were grouped and described by consensus.

Equity, diversity and inclusion statement

Our study focused on professional female football players in Germany. The research team consisted of one woman and seven men. The authors' disciplines included sports medicine, orthopaedics, traumatology and strength, conditioning and exercise science. All the authors were from Germany. We acknowledge that our study excludes semiprofessional and amateur football.

Statistical analysis

Absolute numbers and proportions were reported. The injury mechanism and the role of substitution were analysed with χ^2 tests and Fisher's exact tests. The χ^2 tests were used to compare injury proportions between the two halves of the match and the third part of each half. Cohen's measures and κ measures were used to analyse inter-rater reliability for the observation form. 95% CI are reported. The significance level was set at $p<0.05$. All analyses were conducted using IBM SPSS Statistics, V.28.0.

RESULTS

All 37 ACL injuries (100%) sustained during official matches were included in the video analysis (figure 1). Thirty-four injuries were captured by one camera view and three injuries by two camera views. Most injuries (n=31) occurred in national league matches, compared with only three injuries each in the German Cup and the Champions League. Two ACL injuries were sustained by goalkeepers and 35 by field players: 13 defenders, 16 midfielders and 6 forwards. The mean age was 23.4±3.6 years (range: 18–32 years); the mean weight was 61.6±5.8 kg (range: 48–70 kg), and the mean height was 168.6±4.9 cm (range: 158–184 cm).

The concordance analysis of reliability for the final observation form showed substantial concordance (κ=0.61) with very good concordance (κ>0.8) for 14% of the variables, substantial concordance (κ=0.61 to 0.80) for 43% of the variables and

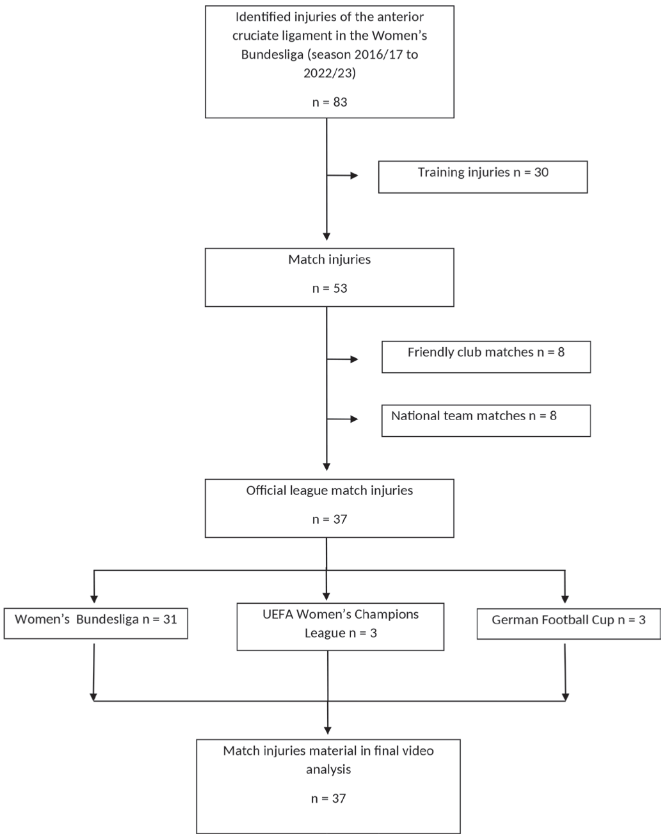

Figure 1 Flow chart. A total of 132 league matches: 33–37 cup matches and 6–10 Champions League matches were played per season.

moderate concordance (κ=0.41–0.60) for 43% of the variables. Concordance was particularly low for the rater's own decision of foul play (0.206).

ACL-associated knee injury

Of the 37 ACL injuries, 21 (57%) were first-time index injuries and 16 (43%) were secondary injuries. Of the 16 secondary injuries, nine (56%) were secondary ipsilateral (to the same side) graft ruptures, and seven (44%) secondary injuries were contralateral ruptures. Two athletes in this cohort sustained their overall third ACL injury. Associated injuries were recorded in 24 athletes (61%). Lateral meniscus injury was noted in 18 athletes (49%), medial meniscus injury in 11 (30%), medial collateral ligament in four (11%) and lateral collateral ligament in one athlete (3%). We found no association between lesion pattern and injury pattern or mechanism.

Contact injury mechanism

The stratified contact mechanisms showed 17 non-contact injuries (46%), 14 indirect contact injuries (38%) and six direct contact injuries (16%).

The contact mechanism did not differ significantly between primary and secondary ACL injuries (injury to the same or contralateral side) (figure 2), but there was a non-significant trend towards a higher non-contact rate in second ACL injuries.

General and football-specific movements

The most common injury situations for field players were horizontal movements (80%). The injury situations for field players are described in detail in table 1:

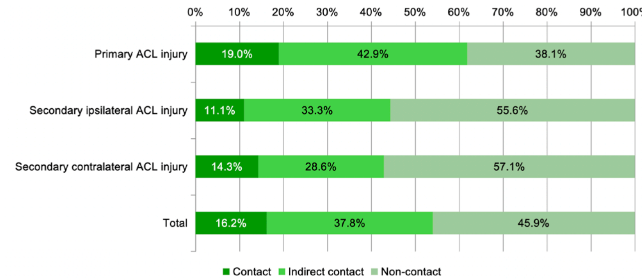

Figure 2 Proportion of contact mechanisms for ACL injury during match play in women's professional football. Contact mechanisms were divided into contact (dark green), indirect contact (light green) and non-contact (very light green). Of the 37 ACL injuries, 21 were first-time injuries, nine were secondary ipsilateral (to the same side), and seven were secondary contralateral injuries. ACL, anterior cruciate ligament.

Four recurrent patterns of ACL match injuries in women's professional football were identified. The patterns differed significantly in their mechanisms and match characteristics: (1) non-contact 'pressing ACL injury' (n=9), (2) indirect contact 'parallel sprinting and tackling ACL injury' (n=7), (3) direct contact 'knee-to-knee ACL injury' (n=6) and (4) non-contact 'landing ACL injury' (n=4). (table 2, figures 3–6).

Two goalkeeper injury situations were identified: (1) the goalkeeper running out of the penalty area to jump shot and clear a ball in front of an attacker. While landing, she had indirect contact with the attacker, who collided with her trunk, which resulted in twisting her knee and (2) after a corner kick, the goalkeeper watched the ball fly across the goal area and out of bounds. While sidestepping, she twisted her knee without any contact.

Field area and foul play

The comparison of the two pitch halves for field players showed no significant differences between the defender's half (54%; 95% CI 36 to 71) and the forward's half (46%; 95% CI 29 to 63, p=0.494). When each pitch half was subdivided into seven areas, the defensive area between the penalty area and the centre line had the highest percentage of ACL injuries in women's professional football (n=9, 24%). Foul play was rarely called by the referee (n=6, 16%) and only in cases of direct contact (n=4, 11%) or indirect contact (n=2, 5%) injuries.

Timing of injury and role of substitution

The two match halves did not differ significantly with regard to injury distribution. The first half (35%; 95% CI 20 to 53) showed a lower proportion of injuries than the second half (65%; 95% CI 47 to 80, p=0.0109). When the match time was divided into six 15-min periods and substitution was taken into account, the first period (0–15 min) showed the highest number of ACL injuries with significant differences to the second period (16–30 min) (p=0.0116) (figure 7A,B). Of the 17 non-contact ACL injuries, seven (41%) occurred during the first 15 min of the player's individual match time.

Of the 37 players who sustained an ACL injury, six (16%) had been substituted. Of these six injuries, four (67%) occurred during the first 15 min of the match, and the other two (33%) occurred during the first 30 min after the substitution.

Pitch condition

There was no rain (0%) in any of the observed injury match situations.

Table 1 General and football-specific movements of field players

General horizontal movements	General vertical movements	Football-specific movements
▶ Sprinting (n=9, 26%) ▶ Change-of-direction manoeuvres (n=6, 16%) ▶ Stopping (n=5, 14%) ▶ Lunging (n=5, 14%) ▶ Standing (n=2, 6%) ▶ Other (n=1, 3%)	▶ Landing (n=6, 17%) ▶ Jumping (n=1, 3%)	▶ Running up (to the ball or opponent) (n=12, 34%) ▶ Clearing (n=5, 14%) ▶ Receiving (n=3, 9%) ▶ Shielding (ball or opponent) (n=3, 9%) ▶ Shooting (n=3, 9%) ▶ Tackling (n=2, 6%) ▶ Dribbling (n=2, 6%) ▶ Blocking (shot) (n=2, 6%) ▶ Other (n=3, 9%)

DISCUSSION

The most important finding of this study was the identification of four common patterns of ACL injuries in official matches of the first women's league in Germany: (1) 'pressing' (2) 'parallel sprinting and tackling', (3) 'knee-to-knee' and (4) 'landing'. We found that a high proportion of 80% of ACL injuries occurred during horizontal movements such as sprinting, stopping, change-of-direction manoeuvres and lunging; only a small proportion

Table 2 Most common recurrent patterns of injuries to the anterior cruciate ligament in women's professional football (field players)

Injury pattern	Frequency n/n total (%)	Main mechanism (n/n total)	Ball possession (n/n total)	Timing of injury (n/n total)	Foul play (referee decision) (n/n total)	Tackle: (n/n total)	Additional details (n/n total)	Movement F (n/n total)	
								Basic	Football-specific
'Pressing ACL injury'	9/37 (24%)	Non-contact (9/9)	Opposing team (9/9)	Unspecific (five in the first half, four in the second half)	Never (9/9)	Never (9/9)	Knee twisting (9/9)	Stopping (5/9) or change of direction (4/9)	Running up to ball or opponent (9/9)
'Parallel sprinting and tackling ACL injury'	7/37 (19%)	Indirect contact (7/7)	Ball possession: injured player (3/7), opposing team (3/7), own team (1/7) Injury pitch location: close to the sidelines (7/7)	In the first 35 min of the first half (7/7)	Rare (fouled 1/7)	Always (7/7), tackle by the opponent (6/7) or own tackle (1/7)	Collision with opponent (2/7), hit/push of opponent (2/7) or other interaction with opponent (3/7); contacted body location of the injured player: shoulder (4/7), hip (2/7) or back (1/7)+knee twisting (7/7)	Sprinting (7/7)	Running up (to ball) (4/7), passing (1/7), tackling (1/7) and shielding (ball/opponent) (1/7)
'Knee-to-knee ACL injury'	6/37 (16%)	Direct Contact (6/6)	Injured player (3/6) or opposing team (3/6)	Unspecific	Common (fouled 3/6; own foul 1/6)	Always (6/6), tackle by the opponent (4/6) or own tackling (2/6)	Kicked by opponent against the knee (5/6) or collision with opponent on the knee (1/6)+knee twisting	Standing (2/6), running/sprinting (2/6), starting (1/6) or lunging (1/6)	Receiving (2/6), clearing (2/6), tackling (1/6) or shielding (ball/opponent) (1/6)
'Landing ACL injury'	4/37 (11%)	Non-contact (4/4)	Injured player (2/4), own team (1/4) or opponent team (1/4)	Unspecific (two in the first half, two in the second half)	Never (4/4)	Never (4/4)	Knee twisting (4/4)	Landing (3/4) or change-of-direction-movement during landing (1/4)	Shooting (1/4), dribbling (1/4) or running to ball or opponent (1/4)
Being tackled ACL injury	3/37 (8%)	Indirect contact (3/3)	Ball possession: injured player (3/3)	Substitution injury (2/3)	Rare (1/3)	Always (3/3)	Hit/push of opponent (2/3) or pull of opponent (1/3)	Change-of-direction-movement (1/3) or stopping (2/3)	Shielding (ball/opponent 2/3) or shooting (1/3)
Disrupted lunging ACL injury	2/37 (6%)	Indirect contact (2/2)	Ball possession: opposing team (2/2)	Second half (2/2)	Never (2/2)	Never (2/2)	Blocked shot during lunge with injured side (1/2) or contralateral side (1/2)	Lunging (2/2) and blocking shot (1/2) or clearing (1/2)	Collision with ball (2)
'Mid-intensity change-of-direction/stopping' ACL injury	2/37 (6%)	Non-contact (2/2)	Opposing team (2/2)	First 5 min (1/) or substitution (1/1)	Never (2/2)	Never (2/2)	Knee twisting (2/2)	Change-of-direction-movement (1/1) or stopping (1/1)	Running to opponent (1/1) or running to ball (1/1)
'Jumping ACL injury'	1/37 (3%)	Non-contact (1/1)	Own team (1/1)	Unspecific	Never (1/1)	Never (1/1)	Unclear (1/1)	Jumping (1/1)	Jumping for header (1/1)
'Shooting' ACL injury	1/37 (3%)	Non-contact (1/)	Own team (1/1)	Unspecific	Never (1/1)	Never (1/1)	Knee twisting (1/1)	Sprinting (1/1)	Cross-shooting the ball (1/1)

ACL, anterior cruciate ligament.

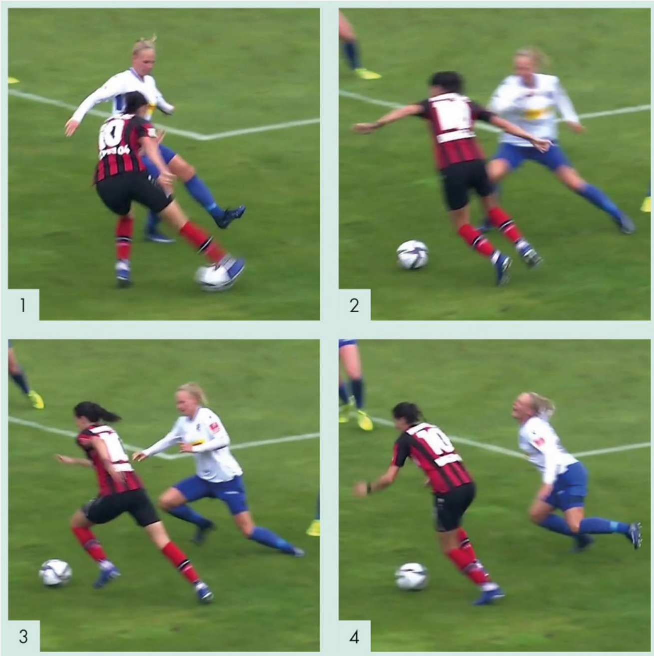

Figure 3 Illustration of the ACL injury pattern in women's professional football: 'pressing ACL injury'. The blue-white player's left knee was injured in this situation. (1) The defending player is running towards the opponent in possession of the ball. (2) She changes direction with a non-contact sidestep cut. (3) Assumed injury frame. (4) Loss of balance. ACL, anterior cruciate ligament.

Figure 5 Illustration of the ACL injury pattern in women's professional football: 'knee-to-knee injury'. The blue-white player's left knee was injured in this situation. (1) Player versus player duel with the defender shielding the ball. (2) She is tackled from behind with a direct lateral impact to the knee joint. (3) Assumed injury frame. (4) Loss of balance. ACL, anterior cruciate ligament.

Figure 4 Illustration of the ACL injury pattern in women's professional football: 'parallel sprinting and tackling knee injury'. The blue-white player's left knee was injured in this situation. (1) Parallel sprinting duel towards the ball. (2) Upper-body tackle and indirect mechanical perturbation while the left leg is on the ground. (3) Assumed injury frame. (4) Loss of balance. ACL, anterior cruciate ligament.

Figure 6 Illustration of the ACL injury pattern in women's professional football: 'landing ACL injury'. The black player's left knee was injured in this situation. (1) Jump over a defender. (2) Non-contact landing with the left leg and visual/neurocognitive perturbation with the head following the ball. (3) Assumed injury frame. (4) Loss of balance. ACL, anterior cruciate ligament.

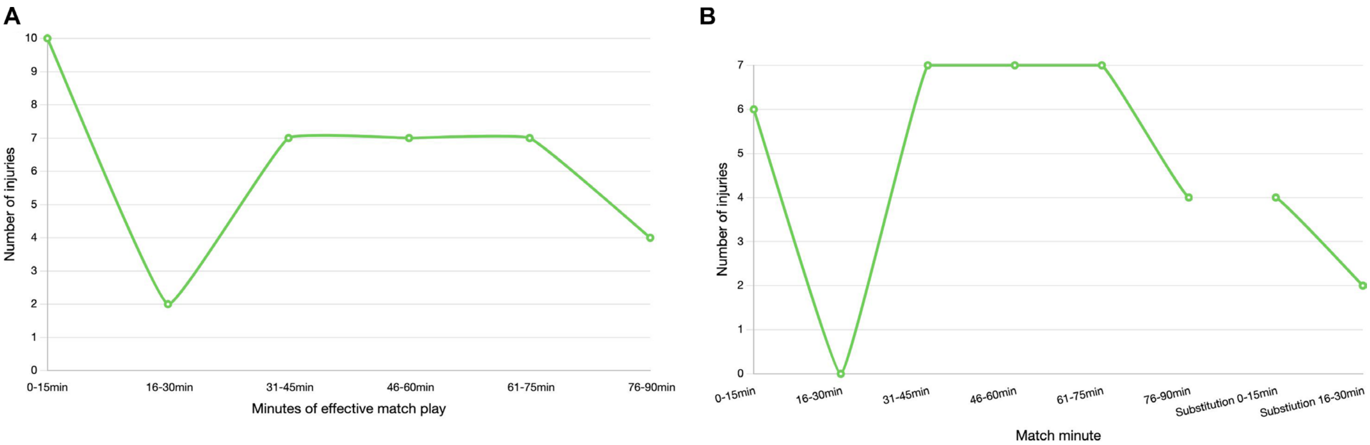

Figure 7 (A) Distribution of match injuries in women's professional football divided by (A) time in effective match minutes and (B) time in minutes.

of injuries resulted from vertical actions (landing and jumping). This finding is consistent with that of previous research,[20] indicating that knee control may be more important during high-risk horizontal movements than during vertical movements.

Contact mechanisms

We found that 84% of ACL injuries in women's professional football occurred without direct contact with the injured knee, which is almost identical to previous findings in women's[19 20] and men's football.[17 18 21] In addition, there was a non-significant trend towards a higher non-contact rate of second ACL injuries on either the same or the contralateral side, which has been described for the first time in women's football. The study also extends findings from ACL-associated knee injuries in women's professional football for the first time.[29 30]

Injury pattern: direct knee-to-knee ACL injury

Football is a sport in which the main aim is to control a ball with the lower extremities and in which physical contact is allowed. Therefore, it is not surprising that direct knee-to-knee contact injury mechanisms were associated with ACL injuries in 16% of situations, which is comparable to previous reports in women's (11%) and men's (12%) professional football.[18 20] In most situations, the injured player was in possession of the ball and the defender was coming mainly from the side or from behind; therefore, the knee contact was probably not visible to the player and thus unexpected. Less often, the injured player was tackling as a defender. Both situations may result in high-impact knee abduction with the leg on the ground.

Foul play was only called in cases of direct knee-to-knee contact injuries. Further research is needed to evaluate how often such tackles lead to severe knee injuries especially when the tackle is not visible to the player or the player is unable to anticipate the knee contact. Injury prevention may focus on increasing awareness and anticipation among the players of these high-risk situations, which may involve better screening of opponents behind the player. However, its effectiveness is unknown.

Injury pattern: parallel sprinting and tackling ACL injury

At 32%, our study reports similar proportions compared with some studies (34%)[20] but lower than other studies (51% tackling injuries).[19] In this indirect contact mechanism, we identified a new common (19%) injury pattern specific to women's football that has not been described before: the parallel sprinting and tackling ACL injury.

'Parallel sprinting and tackling' injuries occur when two opponents sprint in parallel at a seemingly high speed to play the ball rolling away from them. These situations often occur on the sidelines, with the attacker trying to reach the ball at the last moment before it rolls out of bounds. The injured players were at a disadvantage because they were slightly behind the defender but still trying to touch the ball with the foot that was closer to the opponent and with the other (injured) foot firmly on the ground. At that moment, the attacker received an upper-body tackle from the defender shielding the ball. This tackle caused the attacker's torso to move laterally in relation to the stance limb and the knee to move into knee abduction, resulting in an ACL injury.

Lateral perturbation of the trunk is important component of the mechanism during this dynamic high loading of the stance limb before or during ground contact of the injured leg.[31–33] Trunk position and external knee abduction loading may be mechanically related because lateral trunk position may create abduction loads at the knee.[31 32] These frontal plane biomechanics are then further altered by rotation of the upper body while the foot is stationary on the ground, which adds rotational forces to the knee.

This mechanism has a high potential for prevention because trunk and lower extremity control in such high-risk situations may be improved through neuromuscular training.[34 35] Core strength training should specifically target lateral and rotational trunk stability during one-legged stance after sprinting and upper body contact with rotation.[36] In addition, alternative tackling and landing techniques should be taught for these high-risk situations, such as falling backwards on the turf and sliding on the back, and/or rolling backwards over the shoulders after upper-body contact.[37]

Injury pattern: pressing ACL injury

We identified 24% of ACL injuries due to non-contact defensive pressing, which is significantly lower than the percentages reported in the literature on women's (58%–73%)[19 20] and men's (33%–66%)[17 18] professional football. One of the main reasons for this difference is that the other literature reports combined non-contact and indirect contact injuries that occurred during defensive pressing, while we divided these situations into different pattern descriptions.

Injuries occurred during a change-of-direction manoeuvre by the attacker in possession of the ball directly in front of the injured defender and a subsequent—and according to the authors, reactive rather than anticipatory—run towards the ball by the injured defender. Injured players seem to have a high horizontal speed before the injury, followed by a rapid deceleration or change-of-direction manoeuvre in close proximity to the attacker.

Although these specific situations occur regularly in training and competition, it is yet unclear why they lead to immediate knee collapse and subsequent ACL injury. Further research should investigate why such mechanically simple horizontal deceleration manoeuvres lead to match injuries in women's football, whereas male athletes are typically injured during more strenuous jumping manoeuvres.[21 38 39] Due to the close proximity and short reaction time of the defender, a high visual and/or cognitive demand can be proposed as a potential neuromuscular mechanism.[40]

Injury pattern: non-contact landing ACL injury

The fourth most common injury mechanism (11%) is jumping over an opponent and landing on one leg with immediate knee collapse. Most research on predicting ACL injury risk over the past decades has focused on vertical landing differences.[41–44] Our study shows, however, that these injury situations are less common than previously proposed and that research should focus more on understanding horizontal injury mechanisms. For vertical injury prevention, valgus torque seems to be a crucial factor,[41] and proprioceptive deficits in women have been shown to result in quadriceps-dominant activation during landing with delays in hamstring activation that are not seen in men.[44] Women are often affected by the same knee valgus collapse that leads to ACL injury as men but often have significantly higher knee and hip flexion during injury than men.[45]

Time of injury and its relevance for injury prevention

Nearly half of all non-contact injuries occurred during the first 15 min of effective match play. This confirms previous reports in a men's and sex-mixed cohort in professional football of a high proportion of ACL injuries in this early time period[5 18] and extends the findings to the non-contact injury mechanism. Our study (35%) contradicts previous findings in women's professional football (55%)[20] and men's (55%)[18] professional football, which found a higher proportion of ACL injuries in the first half. Timing seems to be an essential factor in non-contact injuries, indicating a high potential for prevention through adequate match preparation and activation. Players and staff members responsible for warm-up may therefore focus more on appropriate warm-up regimens and neuromuscular activation specific to ACL protection[46] as well as on monitoring accumulated fatigue throughout the preseason and season.

Clinical implications

In women's football, there is low-level evidence that multicomponent training programmes reduce ACL injuries by half.[47 48] The identified common injury mechanisms suggest considerable potential for prevention. Multifaceted preventive approaches are needed that focus on the respective basic and sports-specific movement patterns and the most important specific (contact) mechanisms of injury-causing situations.

Strengths and limitations

The strength of this study is its prospective systematic data collection in the highest national women's professional football league in Germany over seven consecutive seasons. To ensure high-validity data, this study used insurance data registered directly by team physicians. This study had a high identification rate of 100%. The study design was identical to that of previous studies, which increases comparability with men's football and other team sports. The main limitation is that most injury situations could only be analysed with a single camera view with limited image quality, especially for injury situations far away from the midline camera position. Systematic bias due to the video analysis by three raters cannot be excluded. The third rater, who was added subsequently, may be biased due to previous results and interpretations. Data from training sessions were not available. Therefore, the injury patterns in training sessions may differ from those in matches. In addition, it is unknown whether the injury mechanisms and playing situations differ from those of amateur or youth football players. This observational study did not include precise analytical methods, such as biomechanical analysis, which adds uncertainty to the reported injury mechanisms.

CONCLUSION

We identified four typical injury patterns leading to ACL injury in women's professional football, which differ significantly in terms of injury mechanisms and match situations. However, most of the common ACL injury patterns identified in women's professional football have great potential for injury prevention.

X Hendrik Bloch @Hendrik_Bloch and Christian Klein @Chrs_Kln

Acknowledgements This study was conducted in cooperation with the German Statutory Accident Insurance for the Administrative Sector VBG and the Deutsches Kreuzbandregister (German ACL registry), who provided the data set on professional female football players.

Contributors LA, DS, WK, CK and HB collected the data and the video footage. TD, CK and LA analysed the video footage. LA, TD, HB and CK analysed the data. All authors, but mainly LA, were responsible for the conception and design of the study. LA, TD, HB and CK interpreted the data. LA wrote the first draft of the paper, which was critically revised by all co-authors. LA is responsible for the overall content as guarantors. All authors have read and approved the final manuscript.

Funding Costs for accessing the video match footage from DieLigen were met by the University of Cooperative Education for Sports and Health, Baunatal, Germany.

Competing interests CK and HB are employed by the VBG.

Patient and public involvement Patients and/or the public were involved in the design, or conduct, or reporting, or dissemination plans of this research. Refer to the Methods section for further details.

Patient consent for publication Not applicable.

Ethics approval Institutional Review Board (IRB) approval was obtained from the Ethics Committee of the University of Regensburg (ID 17-895-101).

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement Data may be obtained from a third party and are not publicly available. Data may have been obtained from a third party and are not publicly available. Epidemiological background data were available from the VBG. These data are not publicly available. The authors have no permission to make the data available. No administrative permissions were required to access the raw data. The VBG prepared and shared the specific files to the University Medical Center Wurzburg. Data protection and confidentiality agreements were signed prior to study research.

ORCID iDs
Leonard Achenbach http://orcid.org/0000-0002-9053-0624
Christian Klein http://orcid.org/0000-0002-8438-5467

REFERENCES

1 López-Valenciano A, Raya-González J, Garcia-Gómez JA, *et al*. Injury profile in women's football: a systematic review and meta-analysis. *Sports Med* 2021;51:423–42.
2 Weitz FK, Sillanpää PJ, Mattila VM. The incidence of paediatric ACL injury is increasing in Finland. *Knee Surg Sports Traumatol Arthrosc* 2020;28:363–8.
3 Beck NA, Lawrence JTR, Nordin JD, *et al*. ACL tears in school-aged children and adolescents over 20 years. *Pediatrics* 2017;139:e20161877.
4 Kooy CEVW, Jakobsen RB, Fenstad AM, *et al*. Major increase in incidence of pediatric ACL Reconstructions from 2005 to 2021: a study from the Norwegian knee ligament register. *Am J Sports Med* 2023;51:2891–9.
5 Waldén M, Hägglund M, Magnusson H, *et al*. Anterior cruciate ligament injury in elite football: a prospective three-cohort study. *Knee Surg Sports Traumatol Arthrosc* 2011;19:11–9.
6 Horan D, Blake C, Hägglund M, *et al*. Injuries in elite-level women's football – a two year prospective study in the Irish women's National League. *Scand J Med Sci Sports* 2022;32:177–90.
7 Hallén A, Tomás R, Ekstrand J, *et al*. UEFA women's elite club injury study: a prospective study on 1527 injuries over four consecutive seasons 2018/19 to 2021/22 reveals thigh muscle injuries to be most common and ACL injuries most burdensome. *Br J Sports Med* 2024;58:128–35.
8 Amundsen R, Thorarinsdottir S, Clarsen B, *et al*. Readytoplay: health problems in women's football – a two-season prospective cohort study in the Norwegian Premier League. *Br J Sports Med* 2024;58:4–10.
9 Larruskain J, Lekue JA, Diaz N, *et al*. A comparison of injuries in elite male and female football players: a five-season prospective study. *Scand J Med Sci Sports* 2018;28:237–45.
10 Waldén M, Hägglund M, Werner J, *et al*. The epidemiology of anterior cruciate ligament injury in football (soccer): a review of the literature from a gender-related perspective. *Knee Surg Sports Traumatol Arthrosc* 2011;19:3–10.
11 Wiggins AJ, Grandhi RK, Schneider DK, *et al*. Risk of secondary injury in younger athletes after anterior cruciate ligament reconstruction: a systematic review and meta-analysis. *Am J Sports Med* 2016;44:1861–76.
12 Arundale AJH, Silvers-Granelli HJ, Snyder-Mackler L. Career length and injury incidence after anterior cruciate ligament reconstruction in major league soccer players. *Orthop J Sports Med* 2018;6:2325967117750825.
13 Lohmander LS, Ostenberg A, Englund M, *et al*. High prevalence of knee osteoarthritis, pain, and functional limitations in female soccer players twelve years after anterior cruciate ligament injury. *Arthritis Rheum* 2004;50:3145–52.
14 Finch C. A new framework for research leading to sports injury prevention. *J Sci Med Sport* 2006;9:3–9.
15 Bahr R, Krosshaug T. Understanding injury mechanisms: a key component of preventing injuries in sport. *Br J Sports Med* 2005;39:324–9.
16 O'Brien J, Finch CF, Pruna R, *et al*. A new model for injury prevention in team sports: the team-sport injury prevention (TIP) cycle. *Science and Medicine in Football* 2019;3:77–80.
17 Waldén M, Krosshaug T, Bjørneboe J, *et al*. Three distinct mechanism predominate in non-contact anterior cruciate ligament injuries in male professional football players: a systematic video analysis of 39 cases. *Br J Sports Med* 2015;49:1452–60.
18 Della Villa F, Buckthorpe M, Grassi A, *et al*. Systematic video analysis of ACL injuries in professional male football (soccer): injury mechanism, situational patterns and biomechanics study on 134 consecutive cases. *Br J Sports Med* 2020;54:1423–32.
19 Brophy RH, Stepan JG, Silvers HJ, *et al*. Defending puts the anterior cruciate ligament at risk during soccer: a gender-based analysis. *Sports Health* 2015;7:244–9.
20 Lucarno S, Zago M, Buckthorpe M, *et al*. Systematic video analysis of anterior cruciate ligament injuries in professional female soccer players. *Am J Sports Med* 2021;49:1794–802.
21 Klein C, Luig P, Henke T, *et al*. Nine typical injury patterns in German professional male football (soccer): a systematic visual Video analysis of 345 match injuries. *Br J Sports Med* 2021;55:390–6.
22 Luig P, Krutsch W, Henke T, *et al*. Contact - but not foul play - dominates injury mechanism in professional men's handball: a video match analysis of 580 injuries. *Br J Sports Med* 2020;54:984–90.
23 Achenbach L, Klein C, Luig P, *et al*. Collision with opponents – but not foul play – dominates injury mechanism in professional men's basketball. *BMC Sports Sci Med Rehabil* 2021;13:94.
24 Waldén M, Mountjoy M, McCall A, *et al*. Football-specific extension of the IOC consensus statement: methods for recording and reporting of epidemiological data on injury and illness in sport 2020. *Br J Sports Med* 2023;57:1341–50.
25 Szymski D, Achenbach L, Zellner J, *et al*. Higher risk of ACL rupture in amateur football compared to professional football: 5-year results of the 'anterior cruciate ligament-registry in German football *Knee Surg Sports Traumatol Arthrosc* 2022;30:1776–85.
26 Szymski D, Koch M, Zeman F, *et al*. Lessons from establishing a football-specific registry of anterior cruciate ligament injuries – data collection and first epidemiological data. *Sci Med Footb* 2022;6:446–51.
27 Gronwald T, Klein C, Hoenig T, *et al*. Hamstring injury patterns in professional male football (soccer): a systematic video analysis of 52 cases. *Br J Sports Med* 2022;56:165–71.
28 Hoenig T, Gronwald T, Hollander K, *et al*. Video analysis of achilles tendon ruptures in professional male football (soccer) reveals underlying injury patterns and provides strategies for injury prevention. *Knee Surg Sports Traumatol Arthrosc* 2023;31:2236–45.
29 Bloch H, Reinsberger C, Klein C, *et al*. High revision arthroscopy rate after ACL reconstruction in men's professional team sports. *Knee Surg Sports Traumatol Arthrosc* 2023;31:142–51.
30 Della Villa F, Hägglund M, Della Villa S, *et al*. High rate of second ACL injury following ACL reconstruction in male professional footballers: an updated longitudinal analysis from 118 players in the UEFA elite club injury study. *Br J Sports Med* 2021;55:1350–6.
31 Hewett TE, Myer GD. The mechanistic connection between the trunk, knee, and anterior cruciate ligament injury. *Exerc Sport Sci Rev* 2011;39:161–6.
32 Hewett TE, Torg JS, Boden BP. Video analysis of trunk and knee motion during non-contact anterior cruciate ligament injury in female athletes: lateral trunk and knee abduction motion are combined components of the injury mechanism. *Br J Sports Med* 2009;43:417–22.
33 Zazulak BT, Hewett TE, Reeves NP, *et al*. Deficits in neuromuscular control of the trunk predict knee injury risk: a prospective biomechanical-epidemiologic study. *Am J Sports Med* 2007;35:1123–30.
34 Paterno MV, Myer GD, Ford KR, *et al*. Neuromuscular training improves single-limb stability in young female athletes. *J Orthop Sports Phys Ther* 2004;34:305–16.
35 Myer GD, Ford KR, Palumbo JP, *et al*. Neuromuscular training improves performance and lower-extremity biomechanics in female athletes. *J Strength Cond Res* 2005;19:51–60.
36 Krutsch W, Lehmann J, Jansen P, *et al*. Prevention of severe knee injuries in men's elite football by implementing specific training modules. *Knee Surg Sports Traumatol Arthrosc* 2020;28:519–27.
37 Achenbach L, Krutsch V, Weber J, *et al*. Neuromuscular exercises prevent severe knee injury in adolescent team handball players. *Knee Surg Sports Traumatol Arthrosc* 2018;26:1901–8.
38 Boden BP, Torg JS, Knowles SB, *et al*. Video analysis of anterior cruciate ligament injury: abnormalities in hip and ankle kinematics. *Am J Sports Med* 2009;37:252–9.
39 Boden BP, Breit I, Sheehan FT. Tibiofemoral alignment: contributing factors to noncontact anterior cruciate ligament injury. *J Bone Joint Surg Am* 2009;91:2381–9.
40 Gokeler A, Tosarelli F, Buckthorpe M, *et al*. Neurocognitive errors and noncontact anterior cruciate ligament injuries in professional male soccer players. *J Athl Train* 2024;59:262–9.
41 Hewett TE, Myer GD, Ford KR, *et al*. Biomechanical measures of neuromuscular control and valgus loading of the knee predict anterior cruciate ligament injury risk in female athletes: a prospective study. *Am J Sports Med* 2005;33:492–501.
42 Quatman CE, Hewett TE. "The anterior Cruciate ligament injury controversy: is "Valgus collapse" a sex-specific mechanism" *Br J Sports Med* 2009;43:328–35.
43 Quatman CE, Quatman-Yates CC, Hewett TE. A 'plane' explanation of anterior cruciate ligament injury mechanisms: a systematic review. *Sports Med* 2010;40:729–46.
44 Hewett TE, Myer GD, Ford KR. Anterior cruciate ligament injuries in female athletes: part 1, mechanisms and risk factors. *Am J Sports Med* 2006;34:299–311.
45 Krosshaug T, Nakamae A, Boden BP, *et al*. Mechanisms of anterior cruciate ligament injury in basketball: video analysis of 39 cases. *Am J Sports Med* 2007;35:359–67.
46 Maniar N, Cole MH, Bryant AL, *et al*. Muscle force contributions to anterior cruciate ligament loading. *Sports Med* 2022;52:1737–50.
47 Crossley KM, Patterson BE, Culvenor AG, *et al*. Making football safer for women: a systematic review and meta-analysis of injury prevention programmes in 11 773 female football (soccer) players. *Br J Sports Med* 2020;54:1089–98.
48 Petushek EJ, Sugimoto D, Stoolmiller M, *et al*. Evidence-based best practice guidelines for preventing anterior cruciate ligament injuries in young female athletes: a systematic review and meta-analysis. *Am J Sports Med* 2019;47:1744–53.